

Exercise 2: SQL Aggregate Functions & Operators:

01. List all distinct departments in the students table:

▶ Run selected (1000) ✓ Just now (11s) workspace.default

```
14
15 --Step 4: Load the information
16 INSERT INTO school.academy.students VALUES
17 (1, 'Alice', 20, 'IT'),
18 (2, 'Bob', 22, 'HR'),
19 (3, 'Charlie', 21, 'IT'),
20 (4, 'Diana', 23, 'Finance'),
21 (5, 'Eve', 22, 'HR');
22
23 --Step 5: Query the table
24 SELECT* FROM school.academy.students;
25
26 --01 List all distinct departments in the students table.
27 SELECT DISTINCT department FROM school.academy.students;
28
29
```

Add parameter

Table +

	department
1	IT
2	HR
3	Finance

3 rows 11.331s runtime See performance (1) ✓ Checked for improvements

Refreshed Now

02. Get the average age of students per department.

▶ Run selected (1000) ✓ Just now (1s) workspace.default

```
17 (1, 'Alice', 20, 'IT'),
18 (2, 'Bob', 22, 'HR'),
19 (3, 'Charlie', 21, 'IT'),
20 (4, 'Diana', 23, 'Finance'),
21 (5, 'Eve', 22, 'HR');
22
23 --Step 5: Query the table
24 SELECT* FROM school.academy.students;
25
26 --01 List all distinct departments in the students table.
27 SELECT DISTINCT department FROM school.academy.students;
28
29 --02 Get the average age of students per department.
30
31 SELECT department, AVG(age) AS avg_age FROM school.academy.students GROUP BY department;
32
```

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	department	avg_age
1	IT	20.5
2	HR	22
3	Finance	23

3 rows 0.821s runtime See performance (1) ✓ Checked for improvements

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03. Show departments with more than 1 student.

Run selected (1000) ✓ 1 minute ago (1s) workspace default

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```
26 --01 List all distinct departments in the students table.
27 SELECT DISTINCT department FROM school.academy.students;
28
29 --02 Get the average age of students per department.
30
31 SELECT department, AVG(age) AS avg_age FROM school.academy.students GROUP BY department;
32
33
34
35 --03 Show departments with more than 1 student.
36
37 SELECT department, COUNT(*) AS student_count
38 FROM school.academy.students
39 GROUP BY department
40 HAVING COUNT(*) > 1;
```

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	department	student_count
1	IT	2
2	HR	2

↓ 2 rows | 0.801s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed 1m ago

04. Get all students whose age is between 21 and 23.

Run selected (1000) ✓ Just now (11s) workspace default

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```
33
34
35 --03 Show departments with more than 1 student.
36
37 SELECT department, COUNT(*) AS student_count
38 FROM school.academy.students
39 GROUP BY department
40 HAVING COUNT(*) > 1;
41
42
43 --04 Get all students whose age is between 21 and 23.
44 SELECT student_id, name, age, department
45 FROM school.academy.students
46 GROUP BY student_id, name, age, department
47 HAVING age BETWEEN 21 AND 23;
48
49
```

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	student_id	name	age	department
1	2	Bob	22	HR
2	3	Charlie	21	IT
3	4	Diana	23	Finance
4	5	Eve	22	HR

↓ 4 rows | 11.392s runtime [See performance \(1\)](#) ✓ Checked for improvements

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05. List all students in the IT or HR department who are older than 21.

Run all (1000) ✓ 1 minute ago (1s) workspace default

47
48
49
50
51 |
52
53
54
55
56
57 --05 List all students in the IT or HR department who are older than 21.
58
59 SELECT*
60 FROM school.academy.students
61 WHERE department IN('IT','HR') AND age >21;
62

Generate (Ctrl + I)

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Table +

	student_id	name	age	department
1	2	Bob	22	HR
2	5	Eve	22	HR

2 rows 0.683s runtime See performance (1) ✓ Checked for improvements

Refreshed 1m ago

06. Show total credits per department, only for departments with more than 5 total credits.

Run selected (1000) ✓ Just now (1s) workspace default

19 (103,'Data science','IT',4),
20 (104,'Excel','Finance',2),
21 (105,'Statistics','HR',3);
22
23 --Step 5: Query the table
24 SELECT* FROM program.catalogue.courses;
25
26
27 --06 Show total credits per department, only for departments with more than 5 total credits.
28
29 SELECT department, SUM(credits) AS total_credits
30 FROM program.catalogue.courses
31 GROUP BY department
32 HAVING SUM(credits)>5;
33
34

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	department	total_credits
1	IT	11

1 row 1.339s runtime See performance (1) ✓ Checked for improvements

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07. List all courses that do not have 4 credits.

▶ Run selected (1000) ✓ Just now (1s) workspace default

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```
25
26
27 --06 Show total credits per department, only for departments with more than 5 total credits.
28
29 SELECT department, SUM(credits) AS total_credits
30 FROM program.catalogue.courses
31 GROUP BY department
32 HAVING SUM(credits)>5;
33
34
35
36 --07 List all courses that do not have 4 credits.
37
38 SELECT* FROM program.catalogue.courses
39 WHERE credits <> 4;
40
```

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	course_id	course_name	department	credits
1	101	SQL Basics	IT	3
2	104	Excel	Finance	2
3	105	Statistics	HR	3

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 3 rows | 0.560s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed Now

08. Show the top 3 courses by credits in descending order.

▶ Run selected (1000) ✓ Just now (1s) workspace default

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```
32 HAVING SUM(credits)>5;
33
34
35
36 --07 List all courses that do not have 4 credits.
37
38 SELECT* FROM program.catalogue.courses
39 WHERE credits <> 4;
40
41 --08 Show the top 3 courses by credits in descending order.
42
43 SELECT* FROM program.catalogue.courses
44 ORDER BY credits DESC
45 LIMIT 3;
46
47
```

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	course_id	course_name	department	credits
1	102	Python	IT	4
2	103	Data science	IT	4
3	101	SQL Basics	IT	3

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 3 rows | 0.942s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed Now

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09. Get the maximum, minimum, and average grade across all enrollments.

```
▶ Run selected (1000) ✓ Just now (1s) workspace default
67 (4,4,104,88),
68 (5,5,105,82);
69
70 --Step 5: Query the table
71 SELECT* FROM test.school.enrollment;
72
73 --09 Get the maximum, minimum, and average grade across all enrollments.
74
75 SELECT MAX(grade) AS max_grade
76 FROM test.school.enrollment;
77
78 SELECT MIN(grade) AS min_grade
79 FROM test.school.enrollment
80
81 SELECT AVG(grade) AS avg_grade
82 FROM test.school.enrollment;
```

Add parameter

Table		
	1 ² max_grade	
1		90

1 row | 1.020s runtime [See performance \(1\)](#) ✓ Checked for improvements

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```
▶ Run selected (1000) ✓ Just now (1s) workspace default
67 (4,4,104,88),
68 (5,5,105,82);
69
70 --Step 5: Query the table
71 SELECT* FROM test.school.enrollment;
72
73 --09 Get the maximum, minimum, and average grade across all enrollments.
74
75 SELECT MAX(grade) AS max_grade
76 FROM test.school.enrollment;
77
78 SELECT MIN(grade) AS min_grade
79 FROM test.school.enrollment;
80
81 SELECT AVG(grade) AS avg_grade
82 FROM test.school.enrollment;
```

Add parameter

Table		
	1 ² min_grade	
1		78

1 row | 0.632s runtime [See performance \(1\)](#) ✓ Checked for improvements

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▶ Run selected (1000) ✓ Just now (1s) workspace: default

67 (4,4,104,88),

68 (5,5,105,82);

69

70 --Step 5: Query the table

71 SELECT* FROM test.school.enrollment;

72

73 --09 Get the maximum, minimum, and average grade across all enrollments.

74

75 SELECT MAX(grade) AS max_grade

76 FROM test.school.enrollment;

77

78 SELECT MIN(grade) AS min_grade

79 FROM test.school.enrollment;

80

81 SELECT AVG(grade) AS avg_grade

82 FROM test.school.enrollment;

Add parameter

Table +

	1.2 avg_grade
1	84.6

Download 1 row 0.778s runtime See performance (1) ✓ Checked for improvements

Refreshed Now

10. Count how many enrollments exist per course.

▶ Run selected (1000) ✓ Just now (<1s) workspace: default

79 FROM test.school.enrollment;

80

81 SELECT AVG(grade) AS avg_grade

82 FROM test.school.enrollment;

83

84

85

86

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89

90 --10 Count how many enrollments exist per course.

91

92 SELECT course_id, COUNT(*) AS enrollment_count

93 FROM test.school.enrollment

94 GROUP BY course_id;

Add parameter

Table +

	1.2 course_id	1.2 enrollment_count
1	101	1
2	102	1
3	103	1
4	104	1
5	105	1

Download 5 rows 0.445s runtime See performance (1) ✓ Checked for improvements

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11. Find total salary and total bonus per department.

▶ Run all (1000) ✓ 04:13 PM (7s) workspace. default

17 INSERT INTO sales.department.salaries VALUES

18 (1,'Tom','IT',60000,5000),

19 (2,'Jerry','HR',55000,4000),

20 (3,'Spike','Finance',70000,6000),

21 (4,'Tyke','IT',62000,5500),

22 (5,'Butch','HR',54000,3500);

23

24 --Step 5: Query the table

25 SELECT* FROM sales.department.salaries;

26

27 --11 Find total salary and total bonus per department.

28

29 SELECT department, SUM(salary) AS total_salary, SUM(bonus) AS total_bonus

30 FROM sales.department.salaries

31 GROUP BY department;

32

Generate (Ctrl + I)

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Table +

123 department

123 total_salary

123 total_bonus

1	IT	122000	10500
2	HR	109000	7500
3	Finance	70000	6000

3 rows 6.750s runtime See performance (1) ✓ Checked for improvements Refreshed 8m ago

12. Show departments where average salary is above 55,000.

▶ Run all (1000) ✓ Just now (<1s) workspace. default

29 SELECT department, SUM(salary) AS total_salary, SUM(bonus) AS total_bonus

30 FROM sales.department.salaries

31 GROUP BY department;

32

33

34

35

36 --12 Show departments where average salary is above 55,000.

37

38 SELECT department, AVG(salary) AS ave_salary

39 FROM sales.department.salaries

40 GROUP BY department

41 HAVING AVG(salary) > 55000;

42

43

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123 department

1.2 ave_salary

1	IT	61000
2	Finance	70000

2 rows 0.395s runtime See performance (1) ✓ Checked for improvements Refreshed Now

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13. List employees whose salary plus bonus is greater than 60,000.

▶ Run selected (1000) ✓ 1 minute ago (13s) workspace default

33

34

35

36 --12 Show departments where average salary is above 55,000.

37

38 SELECT department, AVG(salary) AS ave_salary

39 FROM sales.department.salaries

40 GROUP BY department

41 HAVING AVG(salary) > 55000;

42

43

44 --13 List employees whose salary plus bonus is greater than 60,000.

45 SELECT name, salary, bonus, salary+bonus AS total_income

46 FROM sales.department.salaries

47 WHERE salary+bonus > 60000;

48

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	A name	1 salary	1 bonus	1 total_income
1	Tom	60000	5000	65000
2	Spike	70000	6000	76000
3	Tyke	62000	5500	67500

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3 rows | 13.411s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed Now

14. Show total and average budget per department. Only include departments with average budget above 70,000.

▶ Run all (1000) ✓ 2 minutes ago (2s) workspace default

66 (2,'Payroll system','Finance',80000),

67 (3,'Dashboard','IT',150000),

68 (4,'Website','Marketing',60000),

69 (5,'HR Portal','HR',50000);

70

71 --Step 5: Query the table

72 SELECT* FROM workplace.company.projects;

73

74 --14 Show total and average budget per department. Only include departments with average budget

75 above 70,000.

76

77 SELECT department, SUM(budget) AS total_budget, AVG(budget) AS avg_budget

78 FROM workplace.company.projects

79 GROUP BY department

80 HAVING AVG(budget) > 70000;

81

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	A department	1 total_budget	1 avg_budget
1	IT	270000	135000
2	Finance	80000	80000

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▼

2 rows | 1.525s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed 2m ago

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15. List all projects with budgets between 50,000 and 120,000, excluding the Marketing department.

▶ Run selected (1000) ✓ Just now (1s) workspace default

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74 --14 Show total and average budget per department. Only include departments with average budget

75 above 70,000.

76

77 SELECT department, SUM(budget) AS total_budget, AVG(budget) AS avg_budget

78 FROM workplace.company.projects

79 GROUP BY department

80 HAVING AVG(budget) > 70000;

81

82

83 --15. List all projects with budgets between 50,000 and 120,000, excluding the Marketing department

84 SELECT*

85 FROM workplace.company.projects

86 GROUP BY project_name,project_id,department,budget

87 HAVING department != 'Marketing' AND budget BETWEEN 50000 AND 120000;

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	1 ² project_id	1 ^B project_name	1 ^B department	1 ² budget
1	2	Payroll system	Finance	80000
2	1	AI App	IT	120000
3	5	HR Portal	HR	50000

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3 rows | 0.703s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed Now